

# PAVEL DOLIN

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## SELECTED PUBLICATIONS

### Statistically Valid Post-Deployment Monitoring Should Be Standard for AI-Based Digital Health

P Dolin, W Li, G Dasarathy, V Berisha

Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)    8% acceptance

### ActionReasoningBench: Reasoning about Actions with and without Ramification Constraints

D. Handa<sup>\*</sup>, P Dolin<sup>\*</sup>, S Kumbhar<sup>\*</sup>, C Baral, T Son

13th International Conference on Learning Representations (**ICLR 2025**),    <sup>\*</sup>co-first author

### The Art of Defending: A Systematic Evaluation and Analysis of LLM Defense Strategies on Safety and Over-Defensiveness

N Varshney, P Dolin, A Seth, C Baral

Findings of the Association for Computational Linguistics (**ACL 2024**)

## INDUSTRY EXPERIENCE

### Reddit

June 2026 - Sept 2026

*MLE PhD Intern. Manager: Deguang Kong, PhD*

*San Francisco, CA*

- Identified and solved a novel misalignment between pCTR and pCVR models in Reddit's Core Ads Ranking pipeline. Researched and implemented a KD loss objective aligning pCVR model with pCTR during training and improving CTCVR score during inference by 4.8% PRAUC
- Owned the project end-to-end: designed, trained, and parallelized 30+ experiments to identify and solve the problem, coordinated across multiple teams, shipped the method to production, and authored a research paper on the approach (pending Reddit review and approval).

### Aisera

May 2025 - Aug 2025

*AI/ML Research Intern. Manager: Alexander Liss, PhD*

*Santa Clara, CA*

- Researched and architected a fast inference LLM routing pipeline that reduced baseline serving costs by 50% while maintaining the same accuracy. Led the engineering effort for a production-grade routing pipeline
- Fine-tuned transformer models in PyTorch using PEFT (LoRA/QLoRA, adapters) and selective fine-tuning (layer freezing, last-layer fine-tuning)

### Spiketrapp (NLP startup acquired by Reddit)

Feb 2020 – Aug 2022

*ML Engineer. Manager: Andrea Vattani, PhD*

*San Francisco, CA*

- Received ML Engineer III (IC3) offer from Reddit after Spiketrapp's acquisition; declined to begin PhD at ASU.
- Served as a **lead engineer** for end-to-end development and deployment of listed ML projects:
- Topic Quality Project: Designed, trained, and deployed a topic quality model with custom metrics; built a retraining pipeline for drift handling and integrated into production ML infrastructure.
- Twitter Handles Ads: Designed, trained, and deployed an end-to-end ML pipeline using a lexicon- and embedding-based model to identify semantically similar Twitter handles from seed accounts and their followers
- Text Summarization Project: Researched and evaluated models for summarizing user reviews, balancing cost and quality to select optimal architectures. Built a cost-efficient end-to-end pipeline and owned the full lifecycle.

### Research

- Lead research on Twitch chat sentiment detection and neologisms (emotes) understanding. Created an unsupervised framework, outperforming the previous benchmark by 7.36% points. First author publication.
- Lead research on Twitch "Hateful Raids" (cyberharassment phenomenon) detection and resolution. Successfully created a dataset and an ML + rule-based classifier for raid detections and hate bot/user flagging

## ACADEMIC EXPERIENCE

### ASU ECEE, CHS, Berisha, Dasarathy Groups

August 2024 - Present

*Research Assistant. PIs: Prof. Visar Berisha, Prof. Gautam Dasarathy*

*Tempe, AZ*

- Investigated post-deployment reliability, robustness, and monitoring of AI models in healthcare applications. Proposed a framework for addressing it, leveraging statistical methods familiar to healthcare professionals, researchers, and regulators. First-author publication
- Investigated the theoretical foundations of data scaling laws. Designed a data- and compute-efficient estimation algorithm for learning curves, with applications to generative AI models. First-author publication in progress.

### ASU SCAI, Cognition and Intelligence Group

August 2022 - August 2024

*Research Assistant. PI: Prof. Chitta Baral*

*Tempe, AZ*

- Investigated the capabilities of LLMs on reasoning about actions, effects, and change. Designed and implemented (Python) the experiments. Evaluated and fine-tuned multiple LLM models (LLaMa, Gemma). Analyzed the results of the experiments. Co-first author of the publication. ICLR 2025
- Evaluated, analyzed, and determined the best LLM defense strategies on safety and over-defensiveness. ACL 2024

### UCSB Computational Materials

June 2017 - June 2019

*Undergraduate Researcher. PI: Prof. Anton Van der Ven*

*Santa Barbara, CA*

- First-principles research of thermodynamics of 36 binary refractory alloys. Performed high-throughput quantum mechanical calculations, trained ML models, and ran Monte Carlo to generate phase diagrams. Journal Publication
- Investigated encoding of chemical and structural information of materials for prediction of properties via ML. Conducted research, data analytics, and statistical analysis

## EDUCATION

### Arizona State University

August 2022 – Now

*Ph.D. in Computer Science. Dean's Fellow. GPA: 4.0*

*Tempe, AZ*

*Advised by Visar Berisha & Gautam Dasarathy*

### University of California, Santa Barbara

September 2016 – June 2019

*B.S. in Physics. GPA: 3.77*

*Santa Barbara, CA*

## AWARDS

### NeurIPS 2025 Scholar Award

2025

*Financial support for conference registration and hotel stay*

### Dean's Fellowship ASU

2022 – 2024

*4% of PhD students receive the award. Provides tuition and \$30,000 annually*

### Fulton Fellowship ASU

2022

*An award of \$1000 in "recognition of extraordinary achievements and support of future research and development."*

### Distinction in Major and High Honors UCSB

2019

*Successful defense of honor's thesis and GPA within the 3.50 - 3.79 range*

## TECHNICAL SKILLS

**Languages:** Python, Ruby, C++, BASH, SQL, Matlab

**Packages & Frameworks:** PyTorch, TensorFlow, Scikit-learn, HuggingFace, NLTK, Pandas, NumPy, SciPy, Rails

**Tools & Platforms:** Linux, AWS, Git, Docker, MySQL, PostgreSQL, MongoDB, Redis

**ML & AI:** End-to-end development of predictive models and ML pipelines, large-scale data processing in production systems, training ML and DL models, LLM fine-tuning, PEFT (LoRA/QLoRA, adapters), prompt engineering, reinforcement learning, model evaluation and interpretability, experiment design, A/B testing, expertise in robustness

**Mathematics:** Strong foundation in linear algebra, probability, statistics, optimization, and numerical methods;

expertise in algorithm design, statistical inference, and quantitative problem-solving grounded in first-principles thinking

**Communication & Leadership:** Strong written and verbal communication; experience in teaching and mentoring, cross-functional collaboration, technical leadership, and product management